

Muhammad Shamel

CONTACT INFORMATION

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OBJECTIVE

Passionate IT Infrastructure Architect with expertise in Cloud Computing, IT Infrastructure, and Security. Proven ability to design, implement, and optimize enterprise-grade solutions that enhance efficiency and scalability. Seeking opportunities to drive digital transformation through cutting-edge technologies.

EXPERIENCE

Cairo, Egypt
Aug 2025 – Present

Senior IT Infrastructure & Cloud Architect PwC

Oversee and manage hybrid IT infrastructure across Oracle Cloud (OCI), Microsoft Azure, and on-premises environments. Ensure secure, stable, and optimized infrastructure supporting critical business operations.

Key Responsibilities:

- Manage and maintain OCI and Azure environments, covering compute, storage, networking, load balancers, certificates, IAM, virtual machines, databases, Key Vaults, and Network Security Groups to ensure availability, performance, and security.
- Administer on-premises infrastructure, including Active Directory, DNS, DHCP, Group Policy, and Windows server operations.
- Oversee WSUS patching, system updates, and server maintenance to ensure compliance and minimize vulnerabilities.
- Configure and manage firewalls, VLANs, and routing policies across on-prem and cloud networks for secure and efficient connectivity.
- Manage file services and secure data transfers through SFTP and file server infrastructure.
- Develop and maintain infrastructure documentation, including high-level designs, configuration records, and asset inventories.
- Collaborate with security teams on penetration testing, remediation, and infrastructure hardening initiatives.
- Enhance IT service delivery through ServiceNow catalog development, workflow automation, and process optimization.
- Perform capacity planning, performance monitoring, and continuous improvement to ensure reliability and scalability.

Cairo, Egypt
May 2024 – July 2025

Data Center & Cloud Solution Architect National Bank of Egypt

- Architect and design IT infrastructure to support the bank's business growth, ensuring high availability, scalability, reliability, and security. Oversee the implementation and development of data centers across multiple countries and regions.
- Manage Oracle Cloud Infrastructure (OCI), overseeing compute, storage (block, object, and file), backups, and database management. Responsibilities include identity and access management (IAM), user and group administration, and ensuring robust observability through dashboards and notifications. Also handle network architecture, including VCNs, routing tables, security lists, DRG RPCs, and cross-region connections.
- Network & Capacity Planning – Designing and optimizing network subnets, segmentation, and overall IT infrastructure capacity planning.

- Application Connectivity & Documentation – Building application connection matrices and maintaining comprehensive documentation for all projects.
- Vendor Management & Project Execution – Managing vendors, tracking implementation progress, and ensuring projects meet target deadlines.
- Technical Support & Collaboration – Engaging with support teams to troubleshoot complex technical issues and drive solutions.

Cairo, Egypt
Jul 2022 – April 2024

Network Engineer Orange Business

- Managing and scheduling Network changes for worldwide customers like Henkel, Jaguar, Carl Zeiss, Amcor, DB Schenker, and Takeda.
- Ensuring the professional and successful implementation of customers' Network, Security, and VoIP change management through a comprehensive risk assessment and feasibility analysis

Port said, Egypt
Aug 2020 - Jun 2022

IT Engineer Egyptian Maintenance Company

- Daily monitoring and troubleshooting of Site LAN Network, Microwave links, Telephone network, Public address system, and CCTV system.

CERTIFICATES



Oracle Cloud
Solution Architect
Associate



Microsoft
Azure
Fundamentals



AWS
Cloud Practitioner



CCNA



FortiGate NSE



ITIL

SKILLS

Cloud Platforms: [Oracle Cloud Infrastructure OCI](#), [Azure](#), [AWS](#), [Digital Ocean](#).

Networking & Security: [CCNA](#), [Cisco Meraki](#), [FortiGate](#), [Sophos](#), [Pfsense](#), [VPN](#).

Virtualization & Storage: [Hyper-V](#), [Proxmox](#), [vSphere](#), [Docker](#), [TrueNAS](#), [Block/Object Storage](#).

Operating Systems: [Windows](#), [Linux](#), [Active Directory](#), [DNS](#), [DHCP](#).

Soft Skills: [Problem-solving](#), [Decision-making](#), [self-motivation](#)

EDUCATION

Cairo, Egypt
Aug 2021 - Aug 2022

ICT Postgraduate Diploma National Telecommunication Institute NTI

Focus Areas: Cloud, Networks

GPA: 3.83

Cairo, Egypt
Sep 2013 - Jul 2018

Bachelor of Electronics and Communications Engineering. Arab Academy for Science, Technology and Maritime Transport Grade:

Grade: Very good

EXTRACURRICULARS

Cairo, Egypt
Feb 2017 - Jan 2018

Vice President
AIESEC

Cairo, Egypt
Feb 2016 - Dec 2016

Team Lead
AIESEC

Vadodara, India
Jul 2015 - Aug 2015

Intern
AIESEC

PROJECTS

Dec 2025

OCI Network Security Architecture – Migration to Palo Alto Virtual Firewalls

Description :
Designed and implemented an OCI network security architecture, migrating from OCI native firewalls to Palo Alto virtualized firewalls to enhance security control and optimize operational costs.

Key Responsibilities & Achievements:

- Architected the end-to-end firewall design using Palo Alto VM Series on OCI, including trust, untrust, DMZ zones, and centralized traffic control.
- Designed and implemented OCI networking components, including VCNs, subnets, route tables, and DRG route tables to enforce traffic inspection and segmentation.
- Re-engineered routing flows to steer north-south and east-west traffic through Palo Alto firewalls using custom route tables and DRG attachments.
- Executed migration from OCI native firewall services with minimal downtime and no service disruption.
- Achieved measurable cost savings by consolidating security services while improving visibility and policy control.

May 2024

OCI Data Center Architecture and Implementation – National Bank of Egypt (NBE), KSA

Description :
Designed and delivered a secure, highly available OCI architecture as part of the core architecture team for NBE KSA, aligned with banking regulatory and resilience requirements.

Key Responsibilities & Achievements:

- Architected a multi-site hybrid design spanning two OCI regions in Saudi Arabia, two colocation data centers, and secure remote connectivity to the Riyadh branch and NBE headquarters in Egypt.
- Designed and implemented a hub and spoke network topology enabling centralized security, scalable routing, and controlled inter-network communication.
- Designed and enforced OCI IAM architecture using compartments and least privilege policies to meet security and audit requirements.
- Implemented monitoring and observability using OCI Monitoring, Alarms, and Logging for proactive detection and operational visibility.
- Designed and supported replication and disaster recovery using RackWare to ensure workload portability and DR readiness.
- Worked closely with SAMA for approval, compliance checks, and applications integration

December 2024

Data Center Revamp – NBE NY Branch

Description:

Conducted capacity planning and hardware procurement analysis for the revamp of the Main and DR Data Centers at the National Bank of Egypt – New York Branch, ensuring alignment with business requirements and future scalability.

Key Responsibilities & Achievements:

- Performed **capacity planning analysis** to determine the required hardware specifications based on business modules and workload demands.
- Assisted in **defining infrastructure requirements** for compute, storage, and networking to ensure scalability and efficiency.
- Contributed to the **design and planning phase** for the Data Center revamp, focusing on high availability and disaster recovery readiness.
- **Developed deployment documentation** to standardize implementation processes and ensure long-term maintainability.

Feb 2022 – ongoing

Home lab. Infrastructure Implementation

Description :

Designed and implemented a robust **virtualized infrastructure within my home Lab**, utilizing a high-performance Dell Precision Tower 5810 as the foundation. The project integrated various technologies to enhance computing capabilities, networking, storage, and remote access.

Key Components and Functionalities:

- Utilized **Proxmox** as a bare-metal hypervisor to seamlessly manage multiple virtual machines.
- Configured **Sophos XG Firewall Home Edition** as a firewall and router, establishing secure subnets for VMs and home users, thus enhancing network security.
- Implemented **TrueNAS** for centralized and efficient data storage, ensuring redundancy and easy file sharing.
- Set up **Plex** for seamless media streaming and sharing across devices within the home network.
- Established secure remote access using both **OpenVPN** and **WireGuard** VPN, enabling encrypted connections for managing resources from anywhere.
- Employed **Syncthing** for file synchronization across devices, thereby boosting productivity and collaboration.
- Integrated **Home Assistant** for smart home automation, providing centralized control over IoT devices.
- Deployed **Docker** engines and **Portainer** containers for efficient containerization and management of applications.
- Utilized **AdGuard** for network-wide ad blocking and privacy protection.

Link: https://drive.google.com/file/d/1gW7Qwdstu0o0Ow4pzf94Ym-PRy2DLpG1/view?usp=drive_link
